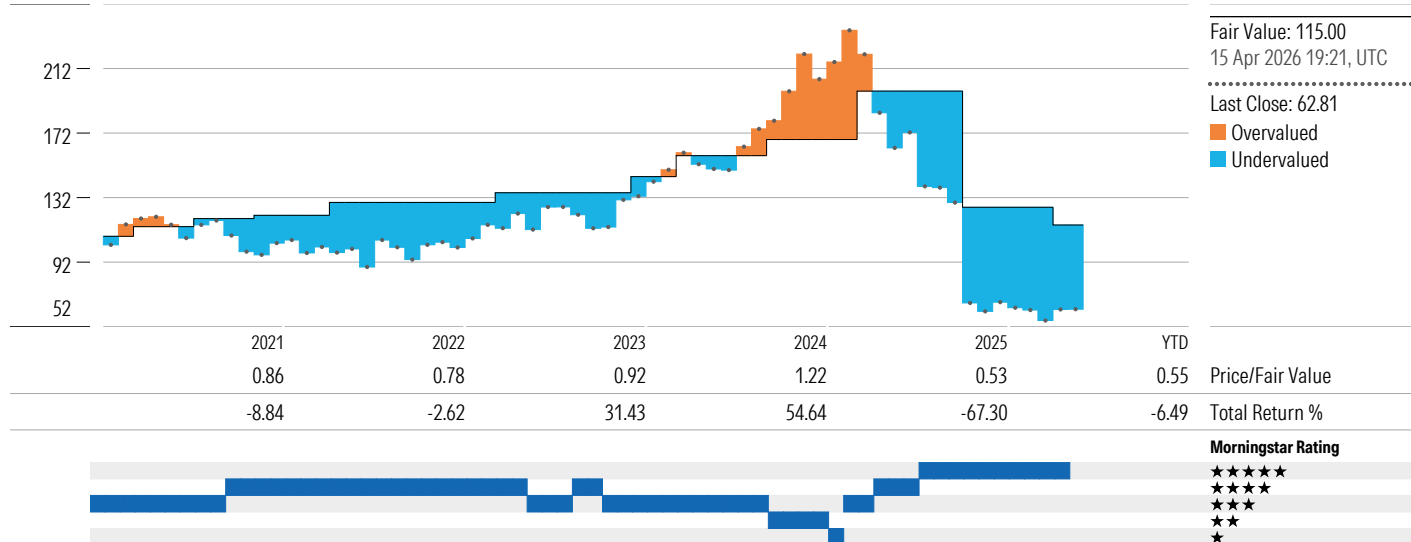


Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
62.81 USD 4 May 2026	115.00 USD 16 Apr 2026 00:21, UTC	0.55	31.20 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

Price vs. Fair Value



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 15 Apr 2026 19:21, UTC.

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Research Methodology for Valuing Companies

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The primary analyst covering this company does not own its stock.

¹The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Fiserv Earnings: Reset Leads to Declines, but Management Maintains Full-Year Guidance

Analyst Note Brett Horn, CFA, Senior Equity Analyst, 5 May 2026

2026 will be a reset year for Fiserv, and management had warned that the first half would be the most difficult. We think the market may have expected better in the first quarter, but the company does appear to be tracking within its guidance.

Why it matters: Fiserv's reset led to a revenue decline in the first quarter and lower margins.

- Organic revenue declined 4% year over year, with the merchant and financials segments down 1% and 6%, respectively. Attrition in the bank tech business remains elevated, but management continues to make changes to address it. Fixing this will likely take some time, but management still expects to return to growth in the back half of the year.
- Adjusted margins declined to 29.7% from 37.8%. Margins will drop further this year as the company reinvests to return to growth. The pain will be strongest in the first half.

The bottom line: We will maintain our \$115 fair value estimate and see the shares as materially undervalued.

- While 2026 will be a difficult year, we believe the company can stabilize the business and return to modest growth over time, as we think it maintains a narrow moat due to scale-based cost advantages in the acquiring business and switching costs on the bank tech side. We are encouraged that management maintained its full-year guidance; we think meeting expectations will be key to

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Sector

Technology

Industry

Information Technology
Services

Business Description

Fiserv is a leading provider of core processing and complementary services, such as electronic funds transfer, payment processing, and loan processing, for US banks and credit unions, with a focus on small and midsize banks. Following its 2019 merger with First Data, Fiserv also provides payment processing services to merchants. About 10% of the company's revenue is generated internationally.

restoring market confidence.

- We are pleased to see Clover generate year-over-year adjusted volume growth of 12% in the quarter, roughly at the midpoint of management's guidance for the year. We see Clover as the key growth engine over time and therefore believe results here will be critical to the company's turnaround efforts.

Business Strategy & Outlook Brett Horn, CFA, Senior Equity Analyst, 15 Apr 2026

Fiserv's merger with First Data in 2019 kicked off a string of three similar deals that took place in short order. While this deal clearly looked like a winner initially, the picture is now more cloudy.

First Data had underperformed its peers prior to the merger, as it was weighed down by an excessive debt load stemming from a leveraged buyout just before the financial crisis and the defection of a major bank partner. We believe this meaningfully limited its ability to reinvest and adapt in an industry that continues to evolve. However, resolving its financial issues put the company back on track. Further, Clover, the company's small-business solution with similarities to Square's offering, has boosted growth, with volume running at an annualized rate of over \$300 billion. Clover offers much stronger pricing and is the type of platform we think the company needs to maintain and build on its leading position in the industry.

While the company's performance over the past few years had been quite strong, new CEO Michael Lyons recently announced that he believes the company has been underinvesting and overly focused on boosting near-term growth. As a result, he intends to reset the business. This will necessitate greater investment and materially lower margins in the near term, along with lower growth. Clover remains the centerpiece of management's growth plans, but even this business will likely see its revenue growth compress a bit. Near-term performance will weaken considerably, and 2026 is likely to be something of a transition year.

The covid pandemic did illustrate one potential negative of the merger: The acquiring business is significantly more macrosensitive than Fiserv's legacy operations. Over the long term, Fiserv's acquiring operations (and Clover specifically) should still be the company's strongest engine for growth. Still, successfully resetting the business could be complicated if the macroenvironment sours before the new CEO can put the company back on a more stable path.

Bulls Say Brett Horn, CFA, Senior Equity Analyst, 15 Apr 2026

- The bank technology business is very stable, characterized by high amounts of recurring revenue and long-term contracts.
- The ongoing shift toward electronic payments has created and will continue to create room for acquirers to see solid growth without stealing share from each other.
- First Data's growth had accelerated before the merger as it worked past its financial issues, and the

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Competitors

	Fiserv Inc FISV	Jack Henry & Associates Inc JKHY	Fidelity National Services Inc FIS	Global Payments Inc GPN
Economic Moat	Narrow	Wide	Narrow	Narrow
Currency	USD	USD	USD	USD
Fair Value	115.00 4 May 2026 16:23, UTC	191.00 4 May 2026 16:21, UTC	85.00 4 May 2026 16:23, UTC	131.00 4 May 2026 16:21, UTC
1-Star Price	178.25	257.85	114.75	203.05
5-Star Price	69.00	133.70	59.50	78.60
Assessment	Undervalued 4 May 2026	Undervalued 4 May 2026	Undervalued 4 May 2026	Undervalued 4 May 2026
Morningstar Rating	★★★★★ 4 May 2026 05:00, UTC	★★★★★ 4 May 2026 05:00, UTC	★★★★★ 4 May 2026 05:00, UTC	★★★★★ 4 May 2026 05:00, UTC
Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst
Capital Allocation	Standard	Exemplary	Standard	Standard
Price/Fair Value	0.55	0.80	0.56	0.55
Price/Sales	1.61	4.55	2.29	2.27
Price/Book	1.29	5.05	1.72	0.75
Price/Earning	9.80	22.10	63.75	16.33
Dividend Yield	0.00%	1.54%	3.47%	1.39%
Market Cap	33.48 Bil	11.01 Bil	24.44 Bil	19.76 Bil
52-Week Range	52.91—191.91	144.12—193.39	43.30—82.74	62.45—90.64
Investment Style	Mid Value	Small Blend	Mid Value	Mid Value

business now has access to greater resources under Fiserv's roof.

Bears Say Brett Horn, CFA, Senior Equity Analyst, 15 Apr 2026

- Since it was built through acquisitions, Fiserv has to maintain multiple core processing platforms, which could limit margins and hamper the quality of its offerings.
- Fiserv's bank technology operations are tied to a mature industry and almost entirely domestic, which limits its growth prospects.
- Due to the First Data merger, Fiserv is now more exposed to macroeconomic conditions.

Economic Moat Brett Horn, CFA, Senior Equity Analyst, 29 Oct 2025

While we believe certain aspects of Fiserv's business benefit from a wide moat, the company's operations have expanded to a point where we believe a narrow moat rating is appropriate for the business as a whole.

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Fiserv's roots lie in core processing systems, the most basic and mission-critical system for banks, and this business is contained in the company's Financial segment, which makes up almost half of revenue. Core processing is the nuts-and-bolts system that banks need to maintain their deposit and loan accounts and to post daily transactions. Given the integral nature of core processing to their operations, banks very rarely switch systems. Besides the potential for interruptions, converting to a new system would require the banks to retrain employees. Customers typically sign multiyear contracts, and customer retention approaches 99% annually, excluding customers lost because of acquisitions by another bank. Fiserv's leading market share also provides an edge in this scalable business. Fiserv's sticky customer relationships and cost advantages add up to a wide economic moat for this portion of the business, in our opinion.

Fiserv's bank offerings have widened significantly over the years, and services related to payment processing, which are also contained in the financials segment, now outstrip core processing and ancillary services in terms of revenue. We believe that this area of the business also enjoys meaningful switching costs, but that the switching costs are not nearly as high as for core processing. In our opinion, cost advantage is the more meaningful factor in this area. Payment processing of any type tends to be highly scalable, and Fiserv enjoys a leading position in processing debit and credit payments for issuers. However, it does not appear to have a dominant edge in terms of volume, and we believe only a narrow moat surrounds this segment of the business.

With its merger with First Data in 2019, Fiserv expanded into merchant acquiring, and this business makes up about half of revenue. Switching costs are relatively low in this area, and the scale and the resulting cost advantage are the primary factors influencing competitive positioning. As a result, a handful of acquirers have come to dominate the industry over time, and mergers and acquisitions in recent years have further consolidated the space. However, First Data remained a leader and enjoys a sizable market share within the US market and is one of three clear volume leaders. If we include its bank joint ventures, Fiserv has historically been the clear volume leader; however, we believe these joint ventures have been a potential source of weakness from a moat perspective, as the company's scale advantage depends on its bank partners. The joint venture with Chase dissolved over a decade ago, and the joint ventures between Bank of America and Wells Fargo recently dissolved. Excluding bank joint ventures, the volume is on par with that of the other leading players. Overall, we think a narrow moat is appropriate for this part of the business.

Fair Value and Profit Drivers Brett Horn, CFA, Senior Equity Analyst, 15 Apr 2026

We are reducing our fair value estimate for Fiserv to \$115 per share from \$117 due to some modest adjustments to our assumptions. Our fair value estimate equates to 14.3 times our revised 2026 adjusted earnings per share projection.

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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We assume that top-line weakness will persist through 2026, leading to only marginal growth next year. Thereafter, we expect overall revenue growth to recover to about 5% over time. We expect growth in the merchant segment to recover over time to about 6%, and the Financial segment to bounce back to almost 4% growth. In both cases, our long-term growth assumptions are below the company's performance in recent years. The net impact of these assumptions is a 4% adjusted revenue CAGR over the next five years.

We expect operating margins to decline materially in 2026, as management increases investment to maintain growth. Our projections assume adjusted operating margins (which exclude amortization and one-time charges) fall about 5 percentage points from 2024 to 2026. Thereafter, we expect margins to hold at this level. Fiserv saw strong margin expansion in 2023 and 2024, and our assumptions essentially assume that adjusted margins retreat to the level prior to this improvement and hold at about 33%.

Risk and Uncertainty Brett Horn, CFA, Senior Equity Analyst, 2 Apr 2026

Any weaknesses in the banking sector could lead banks to defer technology purchases, hurting Fiserv's top line, and any dramatic changes in the structure of the banking industry could have hard-to-predict consequences for Fiserv. Aging core processing systems and increased needs for system flexibility could lead to higher replacement rates over time and possibly erode the company's moat.

The payment processing industry is evolving, and while the position of the acquirers in the current dominant framework is well established, it is possible that future disruption could lower the profitability the industry can generate or cut the acquirers out altogether. As the company's revenue is directly tied to revenue at its merchant customers, Fiserv's acquiring operations are sensitive to macroeconomic conditions.

We see the company's largest ESG risk as data security. Historically, the industry has experienced significant system breaches, which create event risk. The risk of breaches is likely higher on the acquiring side, but a breach on the bank technology side could have greater consequences.

Finally, management has historically been willing to pursue large-scale acquisitions, which could lead to integration risk and high financial leverage at specific points.

Management's plan to raise investment and lower growth in order to find a more stable base raises serious questions about the company's future performance. Our Morningstar Uncertainty Rating is High as a result.

Capital Allocation Brett Horn, CFA, Senior Equity Analyst, 15 Apr 2026

We assign Fiserv a Morningstar Capital Allocation Rating of Standard. In our opinion, the company's balance sheet is sound, its capital investment decisions are fair, and its capital return strategy is

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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appropriate.

Jeff Yabuki held the CEO position from 2005 to 2020, turning the role over to Frank Bisignano, who had previously headed First Data, following the completion of the 2019 merger. Fiserv has completed over 100 acquisitions throughout its history and has essentially operated as a roll-up. However, Fiserv was willing to pursue larger targets at times and made significant moves to reposition the business. The First Data merger was Yabuki's last big move. This deal was very similar to the FIS/Worldpay and Global Payments/TSYS combinations that also occurred in 2019.

We believe these deals, at least in part, were motivated by a desire to improve online payment offerings, as this area has become too large for traditional acquirers to ignore. These deals all bring together companies with operations on the acquirer and issuer side, which should improve the data available to control fraud, a key issue in serving the online payment space. We see Clover as the clearest example of management's ability to successfully navigate a still-evolving industry and maintain Fiserv's scale-based advantage.

Bisignano decided to join the Trump administration, and Michael Lyons took over as CEO upon his exit in May 2025. Lyons was president of PNC Financial Services and previously worked at Bank of America. His history on the banking side should be helpful, given that the bank tech and issuer processing businesses accounts for about half of revenue and the acquiring business remains relatively reliant on the bank referral channel.

Lyons' new approach and comments call Bisignano's tenure into question, as he has stated that the company was underinvesting and overly focused on maintaining near-term growth. Whether Lyons can successfully reset the company's operations and provide a stable base in order to return to growth will be the key factor in the company's long-term success.

As part of the company's plans, the management team and the board have been shuffled. Lyons has appointed new heads for the company's two operating segments and a new CFO. The company has also announced three new board members starting in 2026. One of those new members, Gordon Nixon, will take over as chairman.

Analyst Notes Archive

Fiserv: Shares Rise on Report of Activist Investor but Impact Unclear Brett Horn, CFA, Senior Equity Analyst, 17 Feb 2026

Fiserv's shares rose at a mid-single-digit level based on a report that activist investor Jana Partners had built a stake in the company. Why it matters: The Wall Street Journal reported that Jana Partners has built an equity stake in the company and has discussed the best path forward with management. It is unclear what, if any, impact Jana will have on Fiserv, as the report suggests that the company supports the new CEO and management's current plans. Dramatic restructuring doesn't appear to be part of

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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Jana's plan, and the investor doesn't intend to push for a separation of the bank tech and acquiring businesses, according to the report. The bottom line: We will maintain our \$126 per share fair value estimate and narrow moat rating on Fiserv and see the company's shares as deeply undervalued from a long-term perspective. While we appreciate the difficult near-term that the company faces, we believe Fiserv can achieve modest growth once it moves past this reset. With the company trading at a 13% free cash flow yield (using 2025 results), we see considerable upside if this occurs. We don't have any major issues with management's current plan, but we think execution, as opposed to strategy, is likely the dominant consideration.

Fiserv Earnings: Fourth-Quarter Results and 2026 Guidance About as Expected Brett Horn, CFA, Senior Equity Analyst, 10 Feb 2026

Fiserv's fourth-quarter results were weak in an absolute sense but came in about as expected. With 2026 poised to be a reset year for the company, management's guidance for the year was roughly in line with our projections. Why it matters: The fourth quarter provided further confirmation of the results the company can achieve in the near term as management looks to reset to a more stable base. Revenue was flat year over year on an organic basis, with 1% growth in the merchant segment offset by a 2% decline in the financial segment. As expected, margins dropped significantly as Fiserv boosts spending to drive long-term growth, with adjusted operating margin declining to 34.9% from 42.9% in the year-ago period. The bottom line: We will maintain our \$126 fair value estimate and see the shares as deeply undervalued from a long-term perspective. While we appreciate the difficult near term that the company faces, we believe Fiserv can achieve modest growth once it moves past this reset. With the company trading at a 13% free cash flow yield (using 2025 results), we see considerable upside if this occurs. We think the market might key on a relatively weak volume result for Clover in the fourth quarter. However, this appears to be due to some temporary issues. Management noted that results started to improve in December, and the company's 2026 guidance suggests underlying volume growth could accelerate.

Fiserv Earnings: Management Slashes Guidance and Points to a Complete Reset Brett Horn, CFA, Senior Equity Analyst, 29 Oct 2025

Fiserv's third-quarter numbers were weak and management pointed to a much more difficult future, sending the shares into a spiral on Oct. 29. Why it matters: The quarter itself was poor, with revenue growth fading and margins down. But management's commentary about the future path of the company was the bigger surprise. Year-over-year organic growth was only 1%, with the merchant segment up 5% and the financial segment down 3%. Adjusted operating margin declined 320 basis points year over year. With a new CEO in place, management reviewed the company's position and determined that Fiserv had been attempting to maximize near-term growth and margins in a way that could not be maintained. The company will now raise investment (which will lower margins) and

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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expects growth to decline. Essentially, management pointed to more quarters like this one ahead, until Fiserv resets to a more stable base. The bottom line: After revising our assumptions, we expect to reduce our \$198 fair value estimate for the narrow-moat company by about 35%. The company lowered guidance for 2025, but we have limited insight into how aggressively it will increase investment. We plan to downgrade our Morningstar Uncertainty Rating to High from Medium as a result. While our previous projections were overly optimistic and there is significant uncertainty on where margins and growth will land as the company resets, the current market price is difficult to justify, in our view, representing a free cash flow yield of a little over 10% using the revised guidance for 2025.

Fiserv Earnings: Growth Remains a Little Soft; Market Reaction Overdone Brett Horn, CFA, Senior Equity Analyst, 23 Jul 2025

The market had a strong negative reaction to continued softness in growth at Fiserv, especially with management deciding to pull revenue guidance down to the bottom end of its previous range. Why it matters: Year-over-year organic revenue growth of 8% in the second quarter was a modest improvement from the first quarter, but growth remains below 2024 levels. We think the market remains focused on results at Clover, which again saw muted volume growth of 8% year over year, in part due to a difficult comparison. Management expects Clover volume growth to improve to 10% in the back half of the year. Despite weaker volume growth, year-over-year revenue growth for Clover remained strong at 30%, due to cross-selling. Management reaffirmed its target of \$3.5 billion in revenue for Clover this year. Adjusted operating margins improved 120 basis points year over year, as a decline in margins in the merchant acquiring business was offset by improvement in the bank tech business, which enjoyed relatively strong growth. The bottom line: We will maintain our \$198 per share fair value estimate for the narrow-moat company and see the shares as materially undervalued following the market reaction to the quarter. Management revised its full-year organic revenue growth target to 10% from a previous range of 10% to 12%, attributing this to a slower ramp up in new product offerings outside of Clover. We do not see this as a major long-term issue. We think the market has overreacted to the slowdown in Clover volume growth this year. From a long-term perspective, we still like Fiserv's positioning. On the merchant acquiring side, the company is one of the scale leaders and also has a strong growth vehicle in Clover. In our view, the bank tech business provides some ballast.

Fiserv: Market Reacts Negatively to CFO's Comments on Clover Growth Brett Horn, CFA, Senior Equity Analyst, 15 May 2025

Fiserv's stock dropped significantly on May 15 in reaction to comments that the company's CFO made about near-term volume growth in the Clover business. Why it matters: Clover has seen strong growth in recent years, and we see it as the primary catalyst for growth within Fiserv's merchant acquiring operations. In the first quarter, Clover's year-over-year volume growth was only 8%, down from 14% in the previous quarter. Management pointed to favorable one-time impacts in 2024 (gateway conversions

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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and an extra day in the quarter) that created a difficult comparison. Excluding these factors, the decline in volume growth was much more modest. CFO Bob Hau said during a conference that the company expects Clover volume growth to hold at about 8% in the second quarter, as difficult comparisons will persist over the next couple of quarters. The bottom line: We will maintain our \$198 fair value estimate for the narrow-moat company. While we will monitor the situation, we think the market reaction today has pushed the stock into undervalued territory. The issues negatively affecting Clover's reported volume growth appear to be temporary. Additionally, we think Clover continues to outperform Square, which suggests its competitive position is not weakening. Despite lower volume growth, revenue growth remained strong at Clover at 27% year over year in the first quarter. We think opportunities to cross-sell additional services will support top-line growth even if volume growth slows. Hau reiterated the company's 2025 revenue guidance for Clover at the conference.

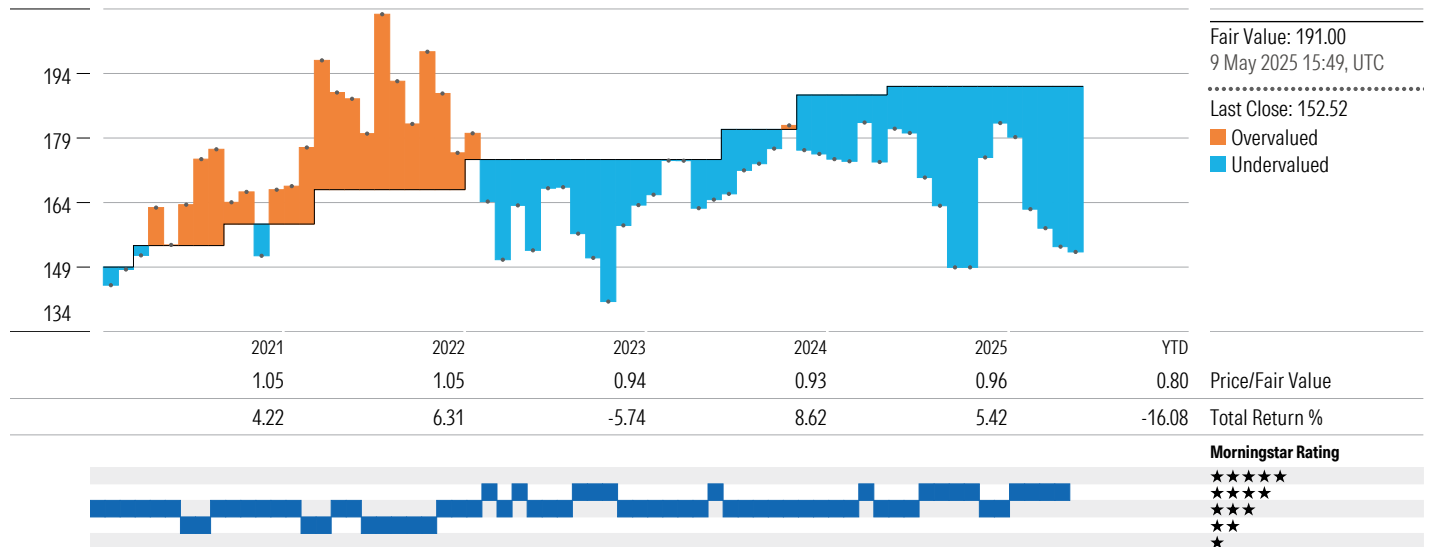
Fiserv Earnings: Slower-Than-Expected Start to the Year Brett Horn, CFA, Senior Equity Analyst, 24 Apr 2025

While management had previously communicated that Fiserv would see a somewhat slow start to the year, the falloff in growth was a little worse than expected, sparking a strong negative reaction from the market. Why it matters: Fiserv's first-quarter results showed a slowdown in growth on the acquiring side, leading to overall year-over-year organic revenue growth of 7%. The merchant solutions segment saw organic revenue growth of only 8%, which marked a significant sequential decline. While this was driven in part by one-time factors (lessened impact of inflation in Argentina, last year being a leap year, and the timing of Easter), underlying growth does appear to be coming back down to earth following strong momentum in 2024. While Clover volume growth slowed significantly to 8%, Clover revenue growth remained strong at 27% due to cross-selling efforts. This dynamic could meaningfully extend Clover's growth, in our view. The bottom line: While the quarter was disappointing, management expects growth to pick up throughout the year, and the company maintained its full-year guidance. Notably, however, the company's guidance assumes stable consumer spending and no large negative impact from tariffs. We think the direction of the macroeconomic environment will ultimately be a bigger factor in full-year results than some softness at the start of the year. However, relative to other payments names, the company's bank tech operations, which account for about half of revenue, provide some ballast in the event of a downturn. We will maintain our \$198 fair value estimate for the narrow-moat company. The market reaction to the quarter will take shares from modestly overvalued to slightly undervalued, in our view. ■■■

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

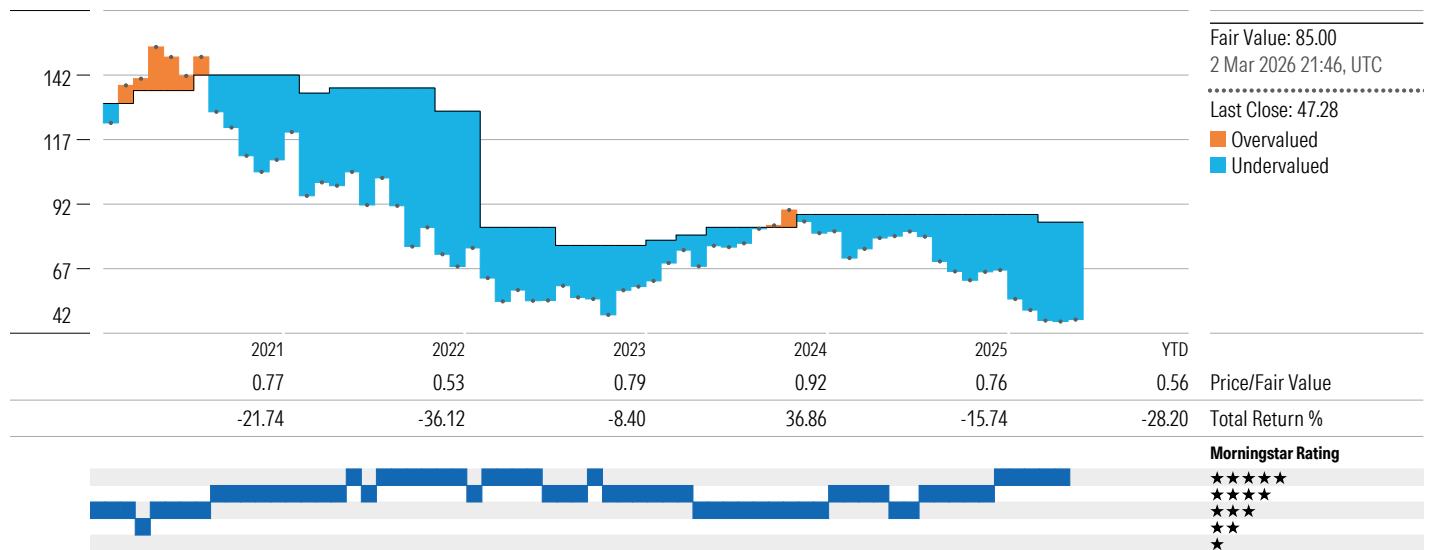
Competitors Price vs. Fair Value

Jack Henry & Associates Inc JKHY



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 9 May 2025 15:49, UTC.

Fidelity National Information Services Inc FIS



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 2 Mar 2026 21:46, UTC.

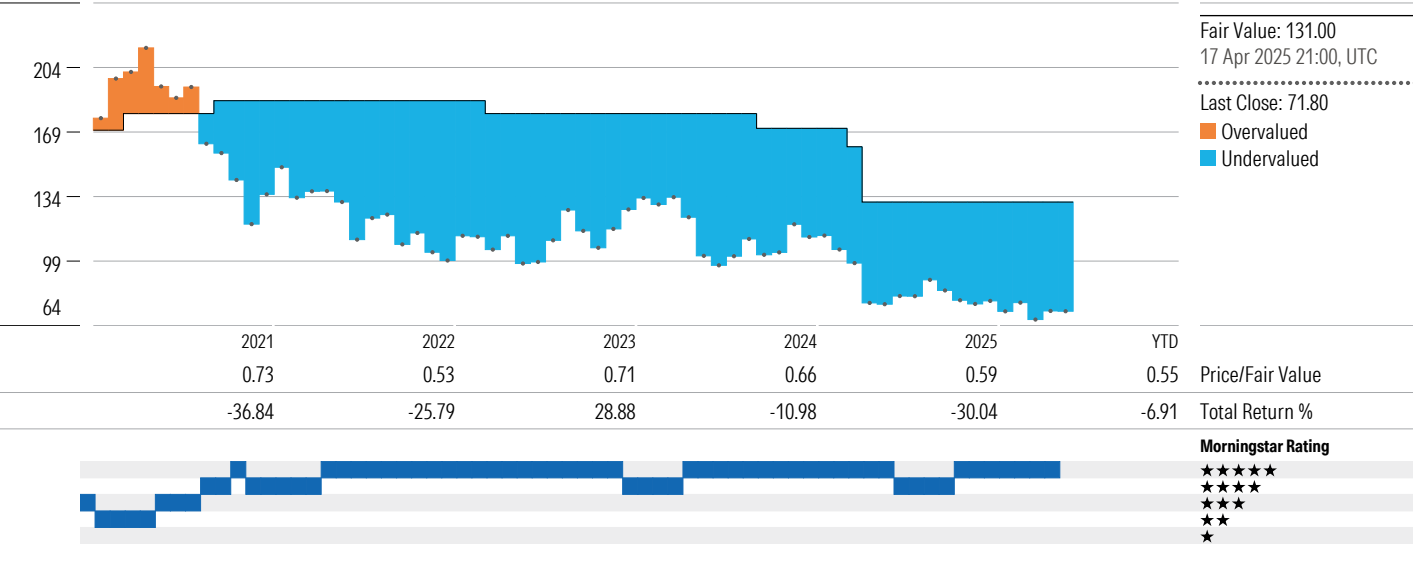
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4 May 2026 21:23, UTC

Competitors Price vs. Fair Value

Global Payments Inc GPN



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Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 17 Apr 2025 21:00, UTC.

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

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Morningstar Valuation Model Summary

Financials as of 15 Apr 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (USD Mil)	19,093	20,456	21,193	21,600	22,425	23,457	24,537	25,684
Operating Income (USD Mil)	4,847	5,879	5,698	5,457	5,797	6,196	6,719	7,133
EBITDA (USD Mil)	7,995	8,796	8,966	8,427	8,679	9,019	9,495	9,902
Adjusted EBITDA (USD Mil)	7,995	8,796	8,966	8,427	8,679	9,019	9,495	9,902
Net Income (USD Mil)	3,068	3,131	3,480	2,992	3,338	3,683	3,904	4,254
Adjusted Net Income (USD Mil)	4,576	4,554	4,688	4,276	4,494	4,723	4,840	5,096
Free Cash Flow To The Firm (USD Mil)	4,822	5,297	5,467	5,171	5,506	5,734	5,970	6,235
Weighted Average Diluted Shares Outstanding (Mil)	616	582	549	532	496	457	419	379
Earnings Per Share (Diluted) (USD)	4.98	5.38	6.34	5.62	6.73	8.05	9.32	11.22
Adjusted Earnings Per Share (Diluted) (USD)	7.43	7.82	8.54	8.03	9.07	10.32	11.56	13.44
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 15 Apr 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	26.9	25.4	28.7	26.9	25.3	25.9	26.4	27.4	27.8	26.5
EBITDA Margin %	—	41.9	43.0	42.3	39.0	38.7	38.5	38.7	38.6	—
Adjusted EBITDA Margin %	—	41.9	43.0	42.3	39.0	38.7	38.5	38.7	38.6	38.7
Net Margin %	15.9	16.1	15.3	16.4	13.9	14.9	15.7	15.9	16.6	15.4
Adjusted Net Margin %	22.8	24.0	22.3	22.1	19.8	20.0	20.1	19.7	19.8	19.9
Free Cash Flow To The Firm Margin %	25.7	25.3	25.9	25.8	23.9	24.6	24.4	24.3	24.3	24.3

Growth & Ratios as of 15 Apr 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	6.1	7.6	7.1	3.6	1.9	3.8	4.6	4.6	4.7	3.9
Operating Income Growth %	15.6	31.5	21.3	-3.1	-4.2	6.2	6.9	8.4	6.2	4.6
EBITDA Growth %	6.6	7.9	10.0	1.9	-6.0	3.0	3.9	5.3	4.3	2.1
Adjusted EBITDA Growth %	6.6	7.9	10.0	1.9	-6.0	3.0	3.9	5.3	4.3	2.0
Earnings Per Share Growth %	17.5	27.6	8.0	17.8	-11.3	19.8	19.5	15.8	20.3	12.1
Adjusted Earnings Per Share Growth %	17.5	11.0	5.3	9.2	-5.9	12.9	13.9	12.0	16.3	12.1

Valuation as of 15 Apr 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earning	17.9	26.3	7.9	7.8	6.9	6.1	5.4	4.7
Price/Sales	4.2	5.7	1.7	1.5	1.5	1.4	1.4	1.3
Price/Book	2.7	4.4	1.4	1.2	1.2	1.1	1.1	1.0
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	12.8	16.0	7.3	7.3	7.1	6.9	6.5	6.2
EV/EBIT	21.1	23.9	11.5	11.3	10.7	10.0	9.2	8.7
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 15 Apr 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	3.4	4.1	4.3	3.8	4.2	4.6	4.8	5.2
ROE %	10.0	11.3	13.5	11.2	12.8	14.3	15.8	17.7
ROIC %	24.6	30.5	28.6	27.0	30.7	35.2	40.8	46.1

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
62.81 USD 4 May 2026	115.00 USD 16 Apr 2026 00:21, UTC	0.55	31.20 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

Financial Leverage (Reporting Currency)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	22.6	17.6	44.6	29.0	28.7	28.6	28.7	29.0
Assets/Equity	3.0	2.8	3.1	3.0	3.1	3.1	3.3	3.4
Net Debt/EBITDA	2.8	2.7	3.2	3.1	3.0	2.8	2.7	2.5
Total Debt/EBITDA	2.9	2.8	3.2	3.1	3.1	2.9	2.8	2.7
EBITDA/ Net Interest Expense	8.2	7.4	6.0	5.8	6.2	6.5	6.0	6.2

Forecast Revisions as of 15 Apr 2026

Prior data as of 2 Apr 2026	2026		2027		2028	
	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	115.00	117.26	—	—	—	—
Revenue (USD Mil)	21,600	21,539	22,425	22,297	23,457	23,197
Operating Income (USD Mil)	5,457	5,491	5,797	5,805	6,196	6,148
EBITDA (USD Mil)	8,427	8,455	8,679	8,678	9,019	8,951
Net Income (USD Mil)	4,276	4,303	4,494	4,500	4,723	4,684
Earnings Per Share (Diluted) (USD)	5.62	5.67	6.73	6.75	8.05	7.97
Adjusted Earnings Per Share (Diluted) (USD)	8.03	8.08	9.07	9.08	10.32	10.24
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 15 Apr 2026

Cost of Equity %	8.9
Pre-Tax Cost of Debt %	5.8
Weighted Average Cost of Capital %	7.8
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	3.0
Stage II Investment Rate %	15.0
Perpetuity Year	15

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 15 Apr 2026

	USD Mil
Present Value Stage I	20,801
Present Value Stage II	28,901
Present Value Stage III	36,110
Total Firm Value	85,813
Cash and Equivalents	798
Debt	28,997
Other Adjustments	0
Equity Value	57,614
Projected Diluted Shares	516
Fair Value per Share (USD)	115.00

Fiserv Inc FISV ★★★★★ 4 May 2026 21:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
62.81 USD 4 May 2026	115.00 USD 16 Apr 2026 00:21, UTC	0.55	31.20 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

ESG Risk Rating Breakdown

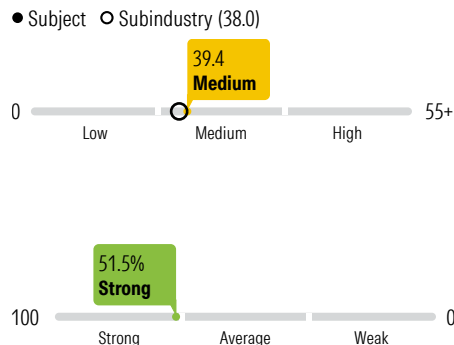
Exposure

Company Exposure¹	39.4
- Manageable Risk	36.2
Unmanageable Risk²	3.2

Management

Manageable Risk	36.2
- Managed Risk³	18.6
Management Gap⁴	17.6

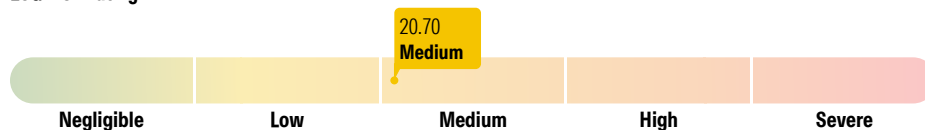
Overall Unmanaged Risk 20.7



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 51.5% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment⁵



ESG Risk Rating is of May 01, 2026. Highest Controversy Level is as of Apr 08, 2026. Sustainalytics Subindustry: Data Processing. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 05 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Fiserv Inc	39.4 Medium	0	55+	51.5 Strong	100	0	20.7 Medium	0	40+
Fidelity Nationa...ion Services Inc	42.0 Medium	0	55+	58.5 Strong	100	0	19.4 Low	0	40+
Global Payments Inc	38.9 Medium	0	55+	56.8 Strong	100	0	18.5 Low	0	40+
Jack Henry & Associates Inc	35.5 Medium	0	55+	41.2 Average	100	0	22.0 Medium	0	40+
—	— —	0	55+	— —	100	0	— —	0	40+

Appendix

Historical Morningstar Rating

Fiserv Inc FISV 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★

Jack Henry & Associates Inc JKHY 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★★	★★★	★★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★	★★	★★	★★	★★	★★★	★★★	★★	★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★	★★★	★★★	★★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★★★

Fidelity National Information Services Inc FIS 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★	★★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★	★★★	★★★	★★★

Global Payments Inc GPN 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our single-point star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a

long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in working capital accounts, and capital spending. Based on these projections, we calculate earnings before interest,

after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



Research Methodology for Valuing Companies

outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

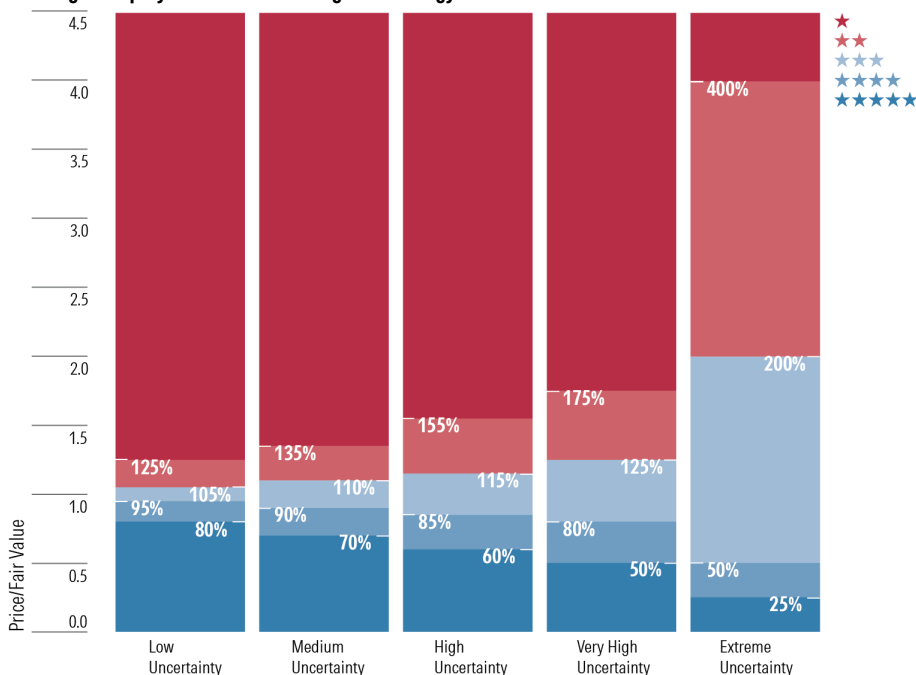
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

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Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

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Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

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